

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Cloud Computing [NPT_4120]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate how the NIST definition of cloud computing supports agility, cost efficiency, and user self-service in modern IT ecosystems. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Evaluate the suitability of hybrid cloud deployment for organizations that need both high security and flexible scalability. (3)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- C) Explain how organizations can integrate IaaS, PaaS, and SaaS models to build a complete cloud-based solution. (2)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Examine how continuous monitoring and automated tools improve SLA compliance in cloud service management. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Compare how utility-based pricing and subscription-based pricing influence cost transparency for cloud consumers. (3)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Explain how performance bottlenecks and tenant isolation issues can affect the efficiency of multi-tenant cloud platforms. (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Examine the consequences of poor data lifecycle management on data security and compliance within cloud storage systems. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Apply the concept of data governance to describe how organizations ensure compliance when storing data across different countries. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Identify which type of storage block, file, or object would be most suitable for hosting large multimedia files and justify your choice. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Evaluate how dynamic resource allocation supports scalability and service continuity in cloud computing environments. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Assess how elastic scaling and intelligent load balancing contribute to handling fluctuating workloads in cloud applications. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Apply monitoring and metering techniques to explain how cloud providers reduce operational costs and power usage. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Evaluate the major security issues and challenges in cloud computing and discuss how they influence trust and data protection across different cloud models. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Explain with examples how cryptography, encryption, and key management are applied to protect data in cloud computing environments. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Examine how Identity and Access Management (IAM) enhances security by controlling user authentication and authorization in cloud systems. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4120:Cloud Computing

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4120:Cloud Computing

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	9(18%)
4	Analysis	11	41(82%)
5	Create	0	0(0%)

6	Evaluate	0	0(0%)
	Total	15	50 (100%)